Ontologies of the LabAutomationTwin

what the lab is, what it can and safely may do, what an experiment is, what the evidence says - and how they connect

Upstream ontologies (AICOR federation, read-only)

- **AICOR L2 ontology** — the hub: scene graph, robots, poses, monitors
- **Robot-scientist vertical** — study designs, interventions, readouts
- **VRB lab-automation-device** — devices and device functions
- **Normative admissibility** — norms, admissible actions
- **DUL / SOMA** — places, concepts, actions

grounded in: 325 concepts placed under 51 upstream classes

Lab ontology

OKF module lab-automation-twin (okf/lab-automation-twin): 413 concepts, 176 properties, 13 modules, HermiT-consistent; extends the upstream ontologies

Why, and with what evidence

culture-experiments
27 concepts, 30 properties
ChallengeRescueStudyDesign
ExperimentalArm
InterventionSpec

records
9 concepts, 6 properties
CellActionRecord
MonitorResult
FailedMonitorResult

experiment-to-motion
34 concepts, 30 properties
CompetingRecoveryModel
GradedRecoveryModel
ThresholdRecoveryModel

How, and safely

tasks
35 concepts, 2 properties
LabTaskType
PlateTransferTask
PlateStoringTask

device-manuals
24 concepts, 23 properties
DeviceManual
CellComponentContract
OperatingProcedure

safe-action
27 concepts, 12 properties
LabHazard
HumanInjuryHazard
CollisionHazard

rsa-kernel
53 concepts, 13 properties
KernelScientificDesignator
KernelLabTaskDesignator
KernelManipulationDesignator

What

devices
73 concepts, 3 properties
PlateStorageFunction
QuarantineFunction
CalibrationFunction

robots
26 concepts, 2 properties
CellRobot
FixedBaseManipulator
RailMountedManipulator

labware
12 concepts, 1 property
Labware
Microplate
SBSMicroplate96

Where

site
63 concepts, 4 properties
AutomatedBiologyLab
LabPremises
LabRoom

core
9 concepts, 50 properties
LabArtifact
LabSDTNode
GroupingNode

poses
21 concepts, 0 properties
TwinPose
KinematicPose
InteractionPose

overlaps culture-experiments: not yet merged

cell01-bridge.ofn (device manuals)

bridge (labware, biology)

prims are instances (ABox)

Experiment ontology

rsa/experiment_ontology.py: 174 concepts, 246 axioms, OWL 2 DL, HermiT

LDL
lab designator language
experiment, challenge_rescue
culture_plan
contracts (4 action types)
processes (4 device processes)
epistemic meta model
experiment_concepts
-> CRAM/RPL plans

Experiments and biology
sections 1-7
Upper level
Relations and partonomy
Biological material and labware
Biological processes
Plans: designs, arms, administrations, comparisons
Factors, levels and treatments
Controls

Designs, evidence, labs
sections 8-13
Readouts and analysis plan
Design classes
Experiments, phases and steps
Data, hypotheses and conclusions
Laboratories, devices and capabilities
Design requirements

capability profiles

Cell 01 package (read-only)

Component records
10 SDT contracts
capabilities
monitors
failure modes
recovery
NEEM record
schemas

Biology ontology
126 classes
cultures, organoids
biological processes
toxic challenge
rescue intervention
hypotheses, predictions

17 alignment subsumptions to Cell 01 biology

Semantic digital twin

USD: the instances of all of these

AICOR Lab 1 and Cell 01 prims
devices, robots, plates, wells
culture SDT layer: belief, interventions, observations
robot episodes, records, images, claims